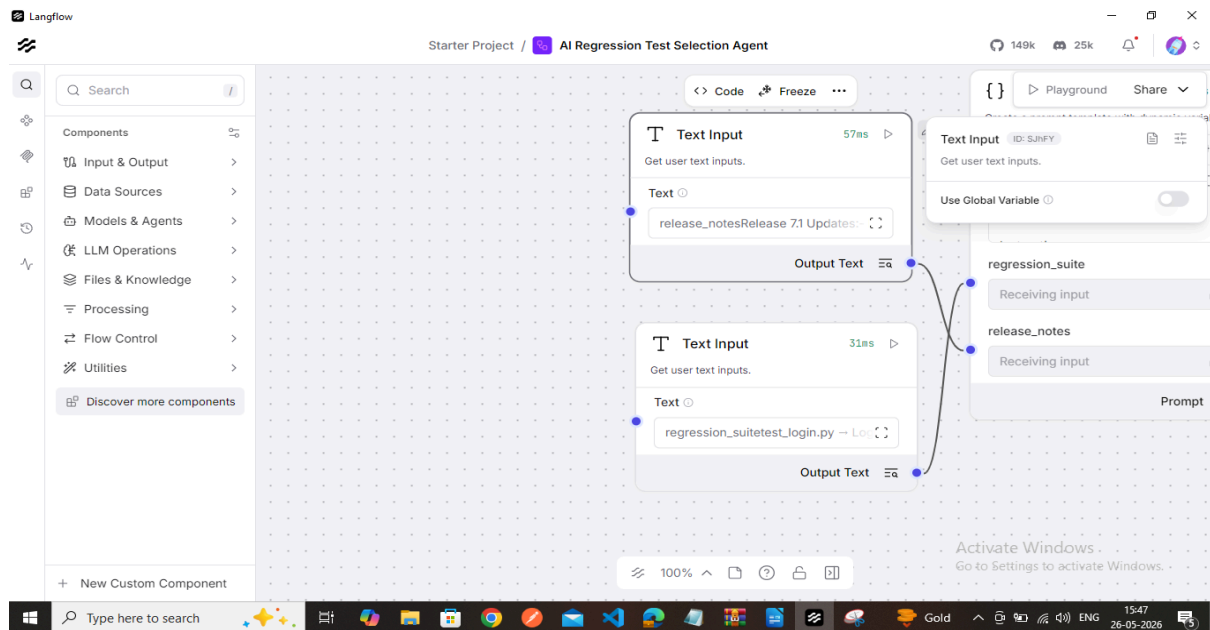
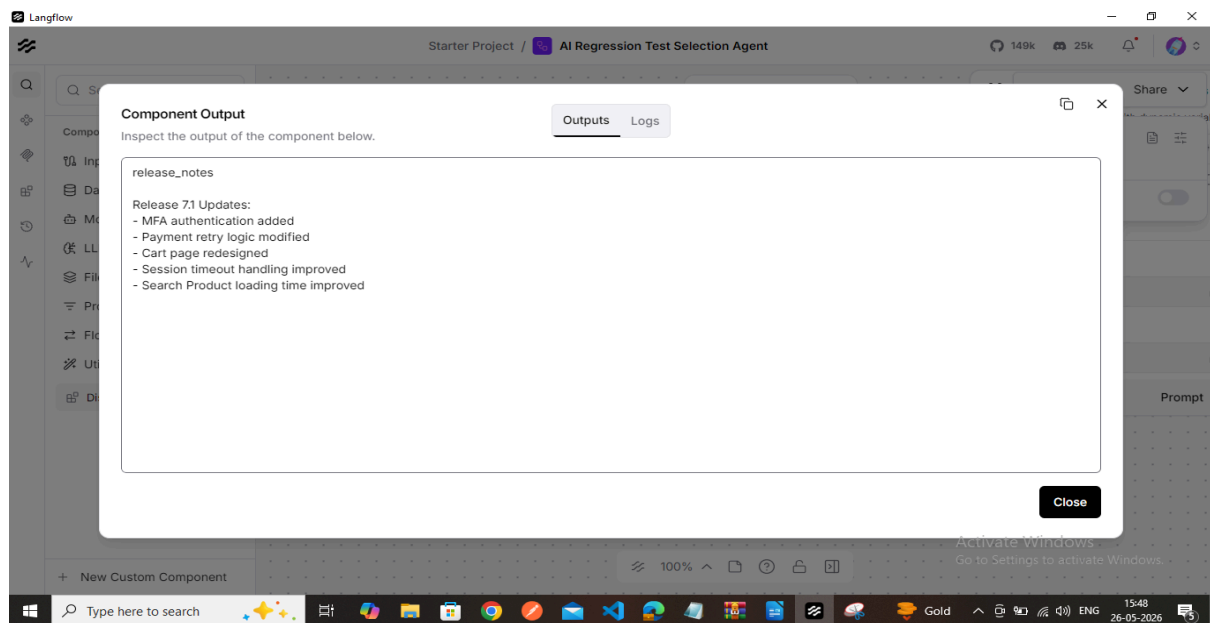


AI Regression Test Selection Agent - POC - Implementation in LangFlow

Added 2 “Text Input” , One for Release Notes and Another for Regression Suite MetaData



Input screenshot for Release notes

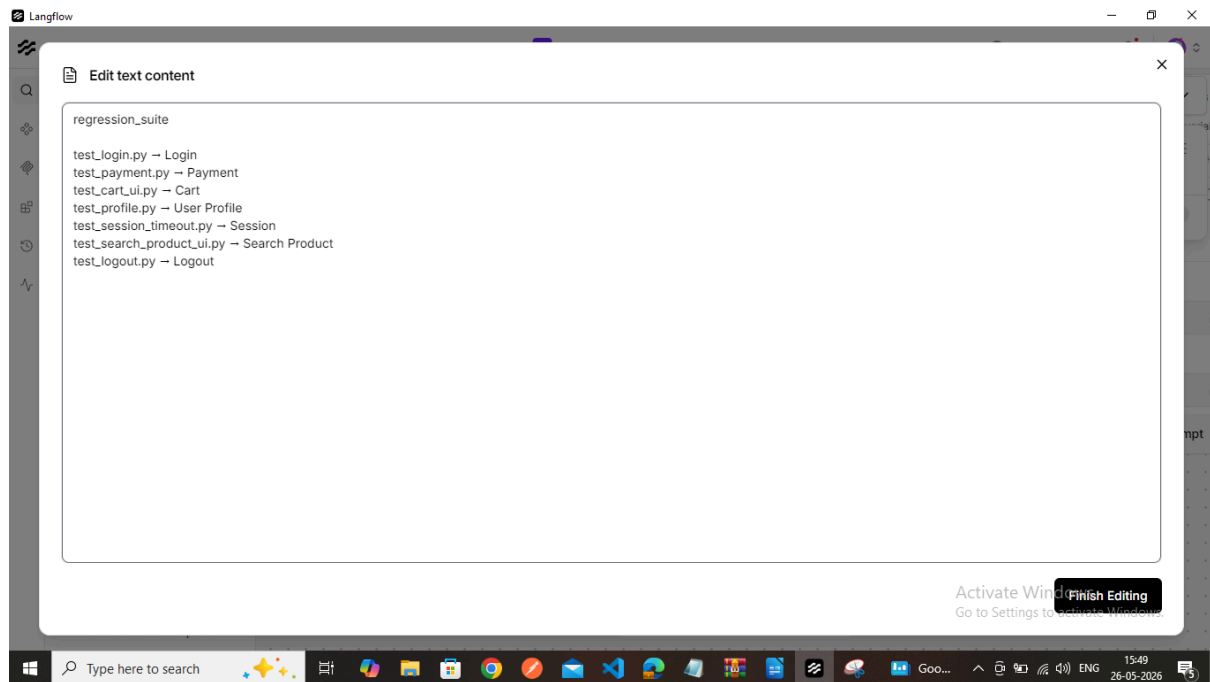


release_notes

Release 7.1 Updates:

- MFA authentication added
- Payment retry logic modified
- Cart page redesigned
- Session timeout handling improved
- Search Product loading time improved

Input screenshot for Regression Suite



regression_suite

test_login.py → Login

test_payment.py → Payment

test_cart_ui.py → Cart

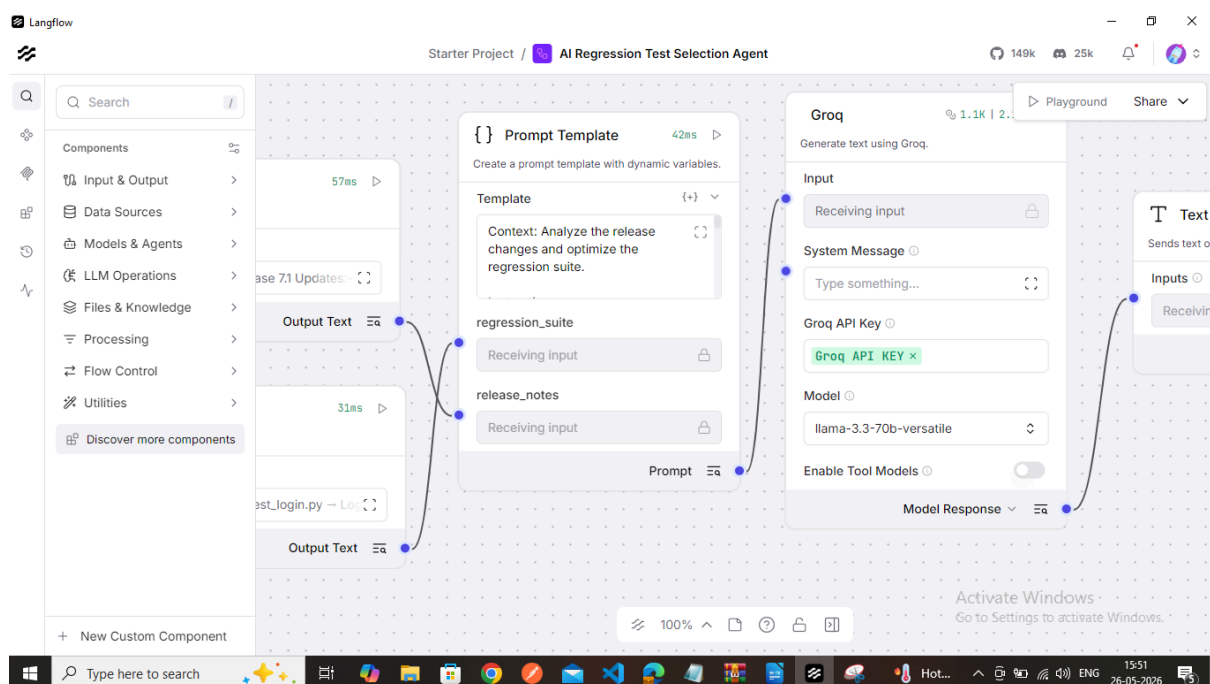
test_profile.py → User Profile

test_session_timeout.py → Session

test_search_product_ui.py → Search Product

test_logout.py → Logout

Added the Prompt Template



Context: Analyze the release changes and optimize the regression suite.

Instruction:

Tasks:

1. Identify impacted modules
2. Select relevant regression scripts
3. Exclude unrelated scripts
4. Highlight risk areas

Rules:

- Select scripts only if related to release changes
- Exclude unrelated modules
- Mention critical risk areas
- Provide clean JSON output

Release Changes:

```
{release_notes}
```

Regression Suite:

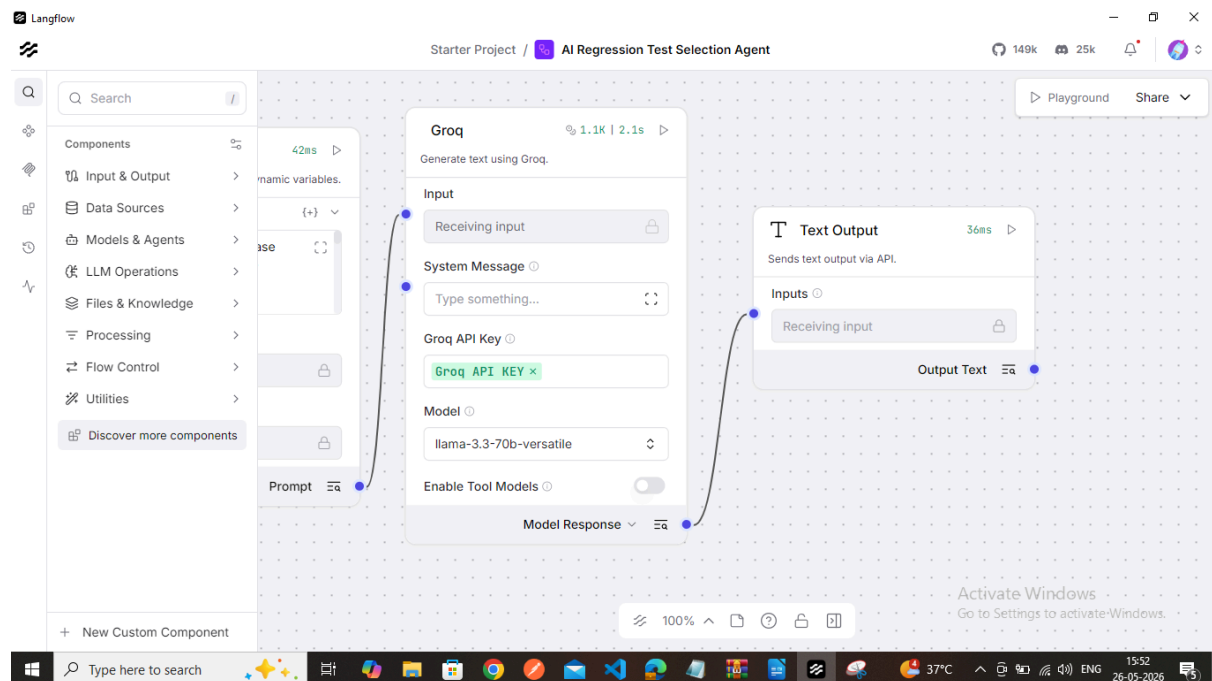
```
{regression_suite}
```

[STRICT] Return Output in JSON file:

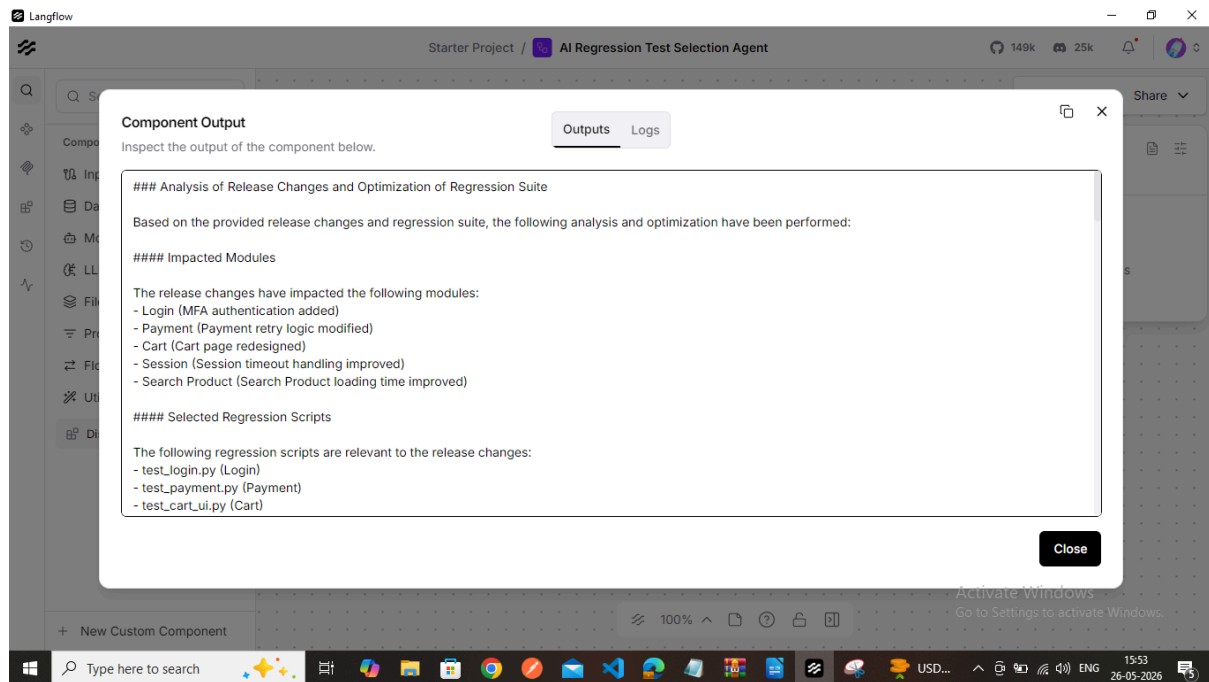
```
{{  
  "selected_scripts": [],  
  "excluded_scripts": [],  
  "risk_areas": []  
}}
```

[STRICT] Return Output in downloadable .html File

Added the LLM as Groq and Model used is **llama-3.3-70b-versatile** and also added Text Output



After running the flow , the output had been generated



Generated Output

Analysis of Release Changes and Optimization of Regression Suite

Based on the provided release changes and regression suite, the following analysis and optimization have been performed:

Impacted Modules

The release changes have impacted the following modules:

- Login (MFA authentication added)
- Payment (Payment retry logic modified)
- Cart (Cart page redesigned)
- Session (Session timeout handling improved)
- Search Product (Search Product loading time improved)

Selected Regression Scripts

The following regression scripts are relevant to the release changes:

- test_login.py (Login)
- test_payment.py (Payment)
- test_cart_ui.py (Cart)
- test_session_timeout.py (Session)
- test_search_product_ui.py (Search Product)

Excluded Scripts

The following scripts are unrelated to the release changes and have been excluded:

- test_profile.py (User Profile)
- test_logout.py (Logout)

Risk Areas

The following areas have been identified as critical risk areas:

- MFA authentication (potential issues with login functionality)
- Payment retry logic (potential issues with payment processing)
- Cart page redesign (potential issues with cart functionality and user experience)
- Session timeout handling (potential issues with session management)
- Search Product loading time (potential issues with search functionality and performance)

JSON Output

```

```json
{
 "selected_scripts": [
 "test_login.py",
 "test_payment.py",
 "test_cart_ui.py",
 "test_session_timeout.py",
 "test_search_product_ui.py"
],
 "excluded_scripts": [
 "test_profile.py",
 "test_logout.py"
],
 "risk_areas": [
 "MFA authentication",
 "Payment retry logic",
 "Cart page redesign",
 "Session timeout handling",
 "Search Product loading time"
]
}
```
```

HTML Output

```
<html>  
<!DOCTYPE html>  
<html>  
<head>  
  <title>Regression Suite Optimization</title>  
  <style>  
    body {  
      font-family: Arial, sans-serif;
```

```

table {
  border-collapse: collapse;
}
th, td {
  border: 1px solid #ddd;
  padding: 10px;
}
</style>
</head>
<body>
  <h1>Regression Suite Optimization</h1>
  <h2>Selected Scripts</h2>
  <ul>
    <li>test_login.py</li>
    <li>test_payment.py</li>
    <li>test_cart_ui.py</li>
    <li>test_session_timeout.py</li>
    <li>test_search_product_ui.py</li>
  </ul>
  <h2>Excluded Scripts</h2>
  <ul>
    <li>test_profile.py</li>
    <li>test_logout.py</li>
  </ul>
  <h2>Risk Areas</h2>
  <ul>
    <li>MFA authentication</li>
    <li>Payment retry logic</li>
    <li>Cart page redesign</li>
    <li>Session timeout handling</li>
    <li>Search Product loading time</li>
  </ul>
  <h2>JSON Output</h2>
  <pre>
{
  "selected_scripts": [
    "test_login.py",
    "test_payment.py",
    "test_cart_ui.py",
    "test_session_timeout.py",
    "test_search_product_ui.py"
  ],
  "excluded_scripts": [
    "test_profile.py",
    "test_logout.py"
  ],
  "risk_areas": [
    "MFA authentication",

```

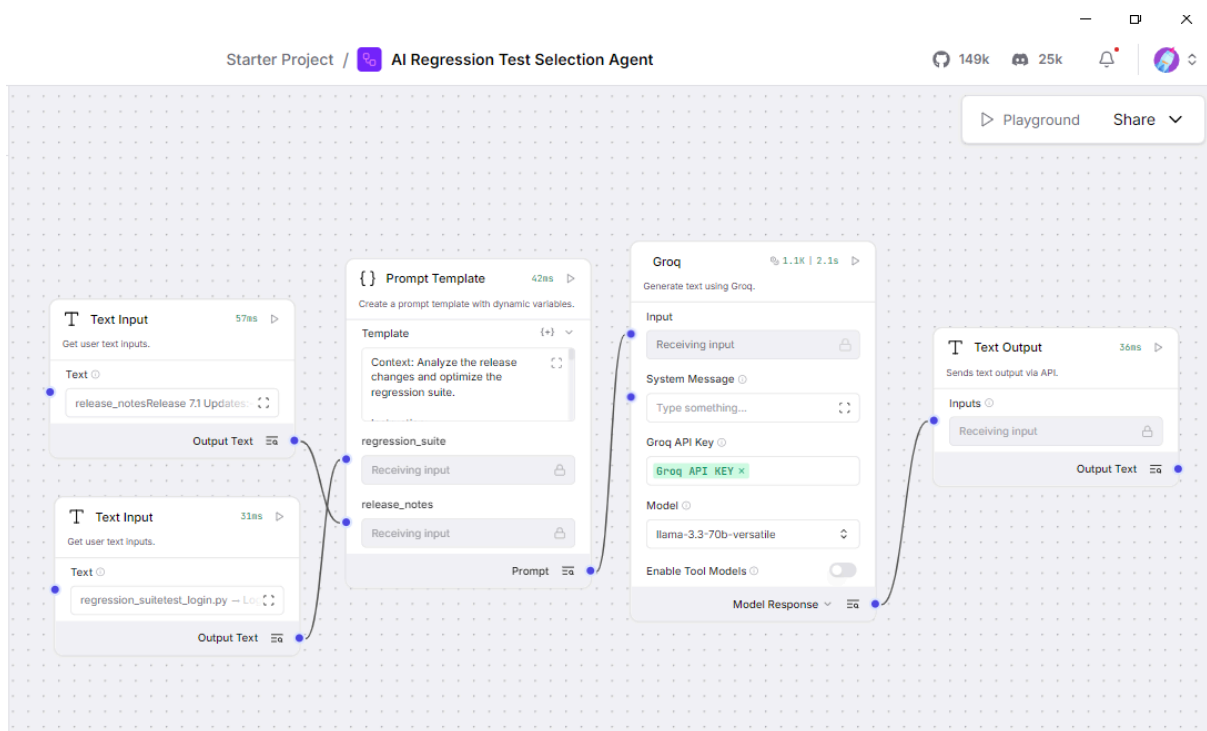
```

    "Payment retry logic",
    "Cart page redesign",
    "Session timeout handling",
    "Search Product loading time"
  ]
}
</pre>
</body>
</html>
'''

```

You can save the above HTML code in a file with a `.html` extension (e.g., `regression\_suite\_optimization.html`) and open it in a web browser to view the output.

Overall Flow Chain



Copied pasted the html code

Regression Suite Optimization

Selected Scripts

- test_login.py
- test_payment.py
- test_cart_ui.py
- test_session_timeout.py
- test_search_product_ui.py

Excluded Scripts

- test_profile.py
- test_logout.py

Risk Areas

- MFA authentication
- Payment retry logic
- Cart page redesign
- Session timeout handling
- Search Product loading time

JSON Output

```
{
  "selected_scripts": [
    "test_login.py",
    "test_payment.py",
    "test_cart_ui.py",
    "test_session_timeout.py",
    "test_search_product_ui.py"
  ],
  "excluded_scripts": [
    "test_profile.py",
    "test_logout.py"
  ],
  "risk_areas": [
    "MFA authentication",
    "Payment retry logic",
    "Cart page redesign",
    "Session timeout handling",
    "Search Product loading time"
  ]
}
```

Activate Windows
Go to Settings to activate Windows.